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Risa 3D

Analysis of a Continuous Beam

Lab 5 – Analysis of a Continuous Beam

Procedure:

In order to complete my lab 5 report I followed the steps given by my instructor through a pdf document. The document contained information in regards to how to create a beam in Risa 3D as well as how to form load combinations, joint members, boundary conditions, along with dead and live loads that should be applied to my continuous beam. All in which are very important in making sure that the structure is stable.

Results:

The following table shows the results of my continuous beam after applying load combinations, joint coordinates, boundary conditions, dead load, live load, and solving via Risa 3D.

Figure 1: Live Load

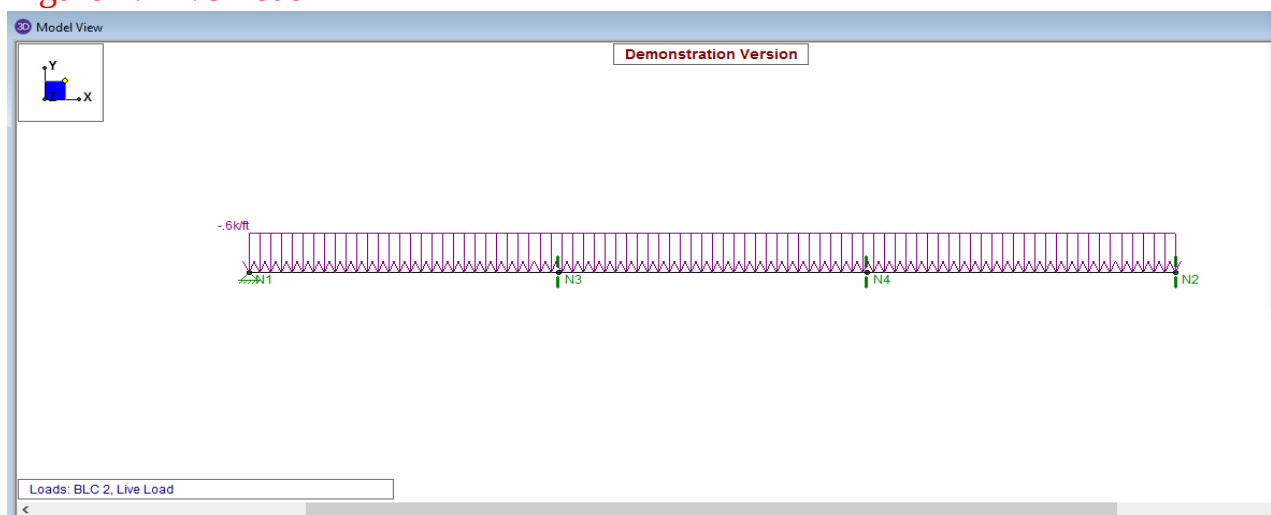


Figure 2: Dead Load

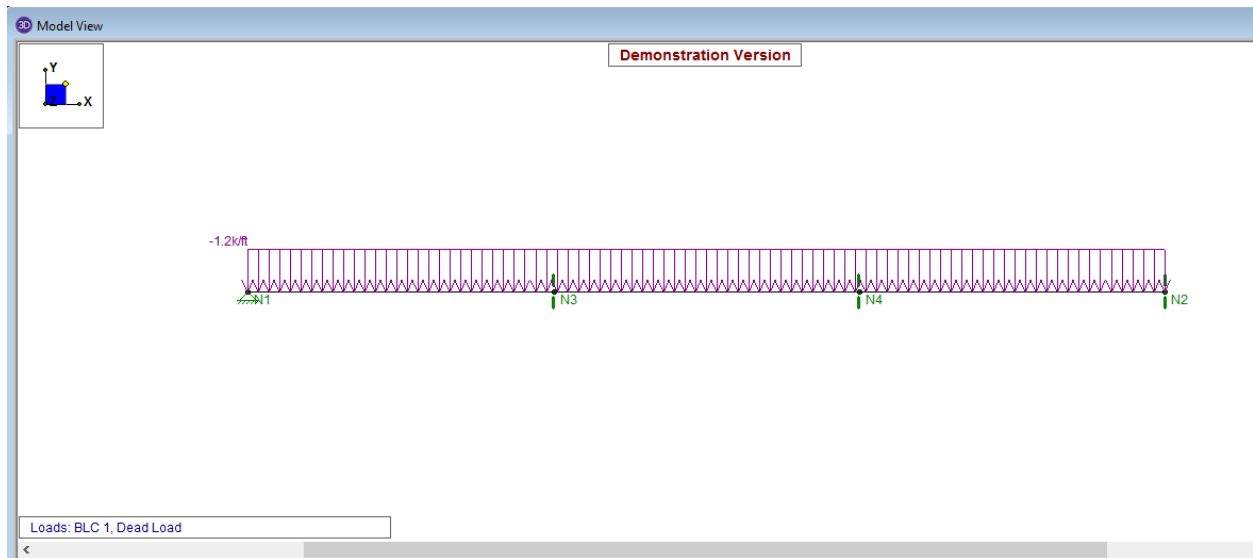


Figure 3: Load Combination

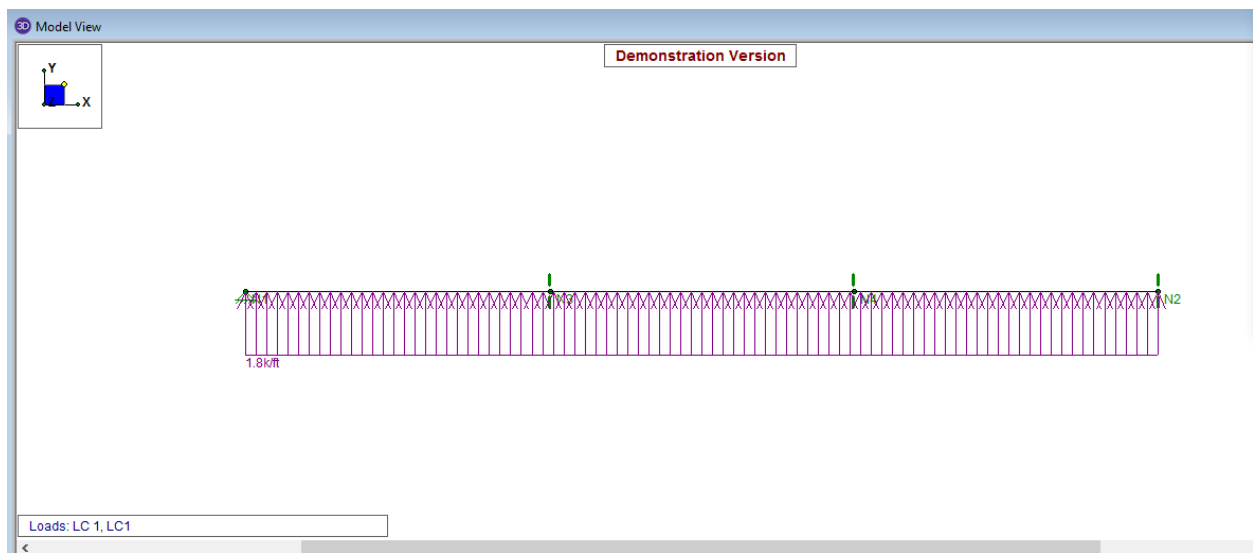


Table 1: Member Forces

Sections Maximums End Reactions									
	L...	Member Label	S...	Axial[k]	y Shea...	z Shea...	Torque[k-ft]	y-y Moment[k-ft]	z-z Moment[k-ft]
1	1	M1	1	0	-18.185	0	0	0	0
2	2		2	0	15.902	0	0	0	15.412
3	3		3	0	0	0	0	0	20.549
4	4		4	0	-15.902	0	0	0	15.412
5	5		5	0	18.185	0	0	0	0

Conclusion:

After applying the data to my continuous beam I was able to easily and efficiently navigate and create the model shown in Figures 1-3 (located in the results section above). After analyzing my results from my Risa 3D data with my handwritten calculations I found a bit of a discrepancy, I believe this is due to the fact that there may be some forces unaccounted for. The best course of action going forward is to go through and analyze each member. The detailed results of my Risa 3D shear and moment output are (Located Below).

